

Towards the Elimination of Mother-to-Child Transmission of HIV in Kenya: A Mathematical and Machine Learning Modeling Study

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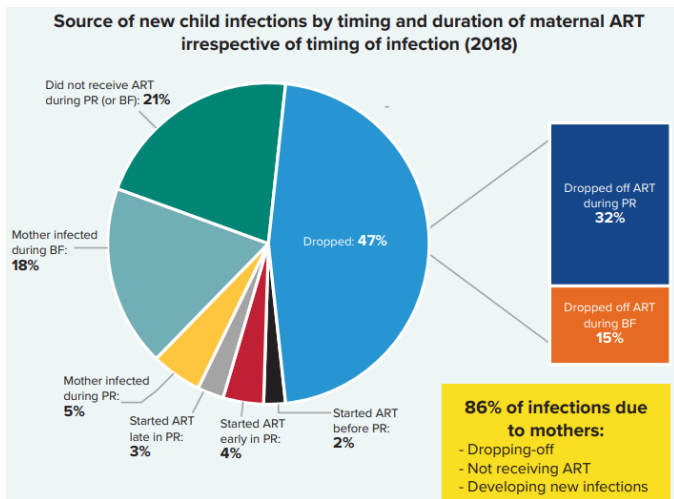
Introduction

Background study

- MTCT of HIV is a significant public health concern with profound biological, health, and social implications.
- Each year, approx 1.3 million women and girls living with HIV become pregnant, with **transmission** rates to their babies ranging from 15% to 45% without intervention (WHO, 2022).
- Kenya has estimated PMTCT coverage of 90% and PMTCT transmission rate of 9.3% (NSDCC 2025).
- HIV infection also has **long-term consequences**, impairing the growth of a child, development, and overall health, impacting their cognitive and physical abilities (Kourtis et al., 2006)

Introduction

Background study: Elizabeth Glaser Pediatric AIDS Foundation (EGPAF). PR = pregnant; BF = breastfeeding; N=7,636



Introduction

Problem statement

- Complex challenges are posed by the PMTCT of HIV in Kenya e.g. high risk of transmission during pregnancy and breastfeeding periods.
- Although women have access to ART, **challenges remain in preventing transmission and maintaining care** (WHO, 2022).
- There is also need to make models calibrated to Kenyan data in order to represent the real dynamics of MTCT of Kenya.
- Machine learning models have also not been extensively applied to predict Kenyan MTCT burden - infection and mortality.

Objectives

General Objective

- To develop and analyze a stochastic and machine learning model for evaluating and optimizing strategies aimed at eliminating mother-to-child transmission of HIV in Kenya.

Specific Objective

1. To formulate a stochastic compartmental mathematical model describing HIV MTCT dynamics under PMTCT interventions.
2. To perform model's mathematical analysis, including stability analysis and well-posedness.
3. To estimate and fit key model parameters using Kenyan epidemiological and PMTCT programmatic data.

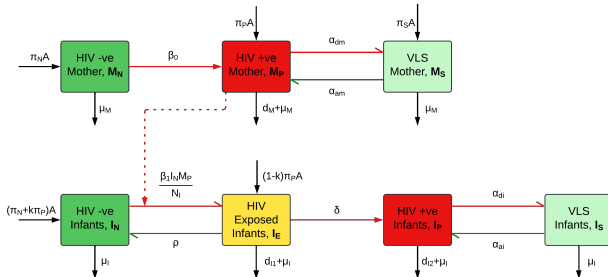
Objectives

Specific Objective

4. To develop an optimal control framework to estimate the impact of prevention strategies such as ART adherence, support services and community-based interventions.
5. To compare machine learning models (logistic regression, KNN, and SVM) for predicting MTCT risk and treatment outcomes using the metrics; precision, accuracy, F1-score, and recall.

Literature Review - mathematical

- A deterministic SEITA model is extended to a stochastic model by incorporating randomness in HIV transmission (Hussain and Alharthi, 2025).
- Maina et. al., 2026 develops a deterministic model for MTCT. Extension proposal: New and known HIV +ve mothers, stochasticity.



Literature Review - machine learning

- To improve HIV care among adolescents; SVM, Logistic Regression, XGBoost, Neural Networks, and Random Forest (Nguyen Tien et. al. 2025).
- Akinsola et al. (2024) applied Random Forest, Logistic Regression, Naïve Bayes, and Decision Tree to predict mother-to-child HIV transmission risk, with Random Forest achieving the best performance.
- Despite widespread PMTCT implementation, Kenya continues to experience MTCT rates above elimination targets due to late ANC attendance, ART interruption, incomplete follow-up, and health system disparities.
- Current approaches lack an integrated data-fitted mathematical model and predictive machine learning framework to accurately predict transmission risk and optimize prevention strategies.

Methods

Proposed flow diagram

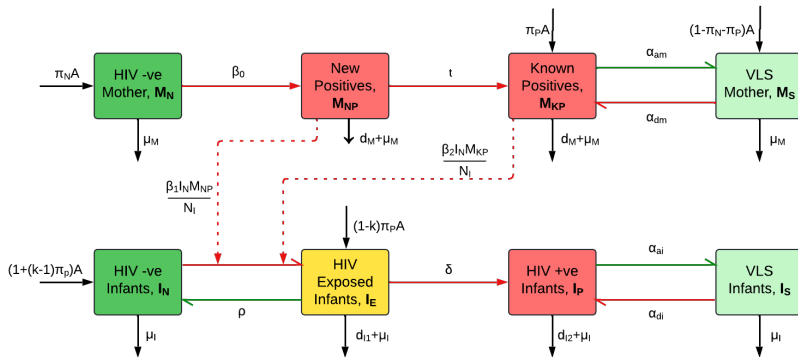


Figure 2: Proposed parameterized compartmental model for Mother to Child transmission

Methods

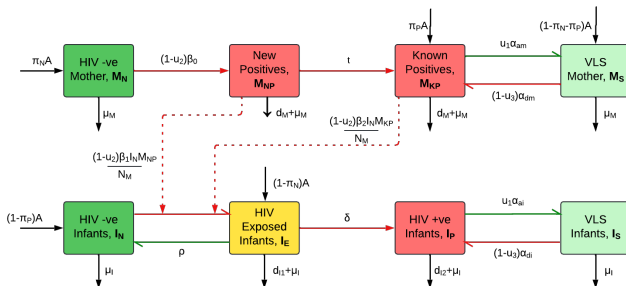
Proposed deterministic model equations

$$\begin{aligned}
 \frac{dM_N}{dt} &= \pi_N A - (\beta_0 + \mu_M) M_N, \\
 \frac{dM_{NP}}{dt} &= \beta_0 M_N - (t + d_M + \mu_M) M_{NP}, \\
 \frac{dM_{KP}}{dt} &= \pi_P A + t M_{NP} + \alpha_{dm} M_S - (\alpha_{am} + d_M + \mu_M) M_{KP}, \\
 \frac{dM_S}{dt} &= (1 - \pi_P - \pi_N) A + \alpha_{am} M_{KP} - (\alpha_{dm} + \mu_M) M_S, \\
 \frac{dI_N}{dt} &= (1 - \pi_P) A + \rho I_E - \left(\frac{\beta_1 M_{NP}}{N_M} + \frac{\beta_2 M_{KP}}{N_M} + \mu_I \right) I_N, \\
 \frac{dI_E}{dt} &= (1 - \pi_N) A + \left(\frac{\beta_1 M_{NP}}{N_M} + \frac{\beta_2 M_{KP}}{N_M} \right) I_N - (\rho + d_{I1} + \delta + \mu_I) I_E, \\
 \frac{dI_P}{dt} &= \delta I_E + \alpha_{di} I_S - (d_{I2} + \alpha_{ai} + \mu_I) I_P, \\
 \frac{dI_S}{dt} &= \alpha_{ai} I_P - (\mu_I + \alpha_{di}) I_S.
 \end{aligned} \tag{1}$$

Methods

Proposed optimal control model

- $u_1(t)$: ART adherence by mothers and infants (α_{am}, α_{ai}),
 $u_2(t)$: Support services: ($\beta_0, \beta_1, \beta_2$) and $u_3(t)$:
 Community-based interventions: (α_{dm}, α_{di}).



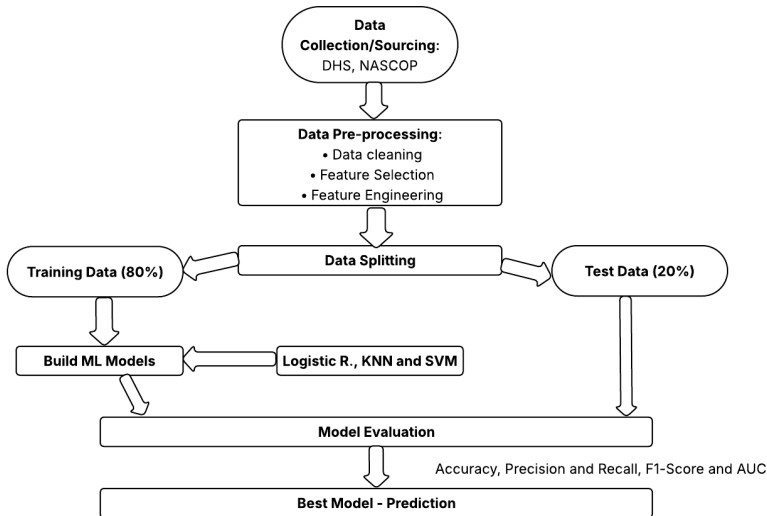
Methods

Proposed stochastic model equations

$$\begin{aligned}
 dM_N &= [\pi_N A - (\beta_0 + \mu_M) M_N] dt - \sigma_1 \beta_0 M_N dB_1, \\
 dM_{NP} &= [\beta_0 M_N - (t + d_M + \mu_M) M_{NP}] dt + \sigma_1 \beta_0 M_N dB_1, \\
 dM_{KP} &= [\pi_P A + t M_{NP} + \alpha_{dm} M_S - (\alpha_{am} + d_M + \mu_M) M_{KP}] dt, \\
 dM_S &= [(1 - \pi_N - \pi_P) A + \alpha_{am} M_{KP} - (\alpha_{dm} + \mu_M) M_S] dt, \\
 dI_N &= \left[(1 - \pi_P) A + \rho I_E - \left(\frac{\beta_1 M_{NP}}{N_M} + \frac{\beta_2 M_{KP}}{N_M} + \mu_I \right) I_N \right] dt \\
 &\quad - \frac{\sigma_2 \beta_1 M_{NP} I_N dB_2 + \sigma_3 \beta_2 M_{KP} I_N dB_3}{N_M}, \\
 dI_E &= \left[(1 - \pi_N) A + \frac{\beta_1 M_{NP}}{N_M} + \frac{\beta_2 M_{KP}}{N_M} - (\rho + \delta + d_{I1} + \mu_I) I_E \right] dt \\
 &\quad - \frac{\sigma_2 \beta_1 M_{NP} I_N dB_2 + \sigma_3 \beta_2 M_{KP} I_N dB_3}{N_M}, \\
 dI_P &= [\delta I_E + \alpha_{di} I_S - (\alpha_{ai} + d_{I2} + \mu_I) I_P] dt, \\
 dI_S &= [\alpha_{ai} I_P - (\alpha_{di} + \mu_I) I_S] dt.
 \end{aligned} \tag{2}$$

Methods

Proposed machine learning framework



Methods

Methods for specific objectives

- **Stability analysis:** Ruorth-Hurtwiz criterion, Metzler matrix method. **Uncertainty and sensitivity analysis:** Latin Hypercube Sampling (LHS) technique. Fitting using MCMC method.
- Data Nature; **Target:** Infant diagnosis(HIV +ve or -ve) ; **Predictor variables:** Viral load count during pregnancy, ART regimen, coinfection, parity, mode of delivery, place of delivery(facility/home), skilled birth attendance, infant birth weight prematurity status, No. of ANC visits, marital status, partner HIV/ART status.
- Optimal control: Pontryagin's Maximum Principle - ART adherence (α_{am}, α_{ai}), support services to prevent dropout ($\beta_0, \beta_1, \beta_2$), and community-based approaches (α_{dm}, α_{di})

References

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